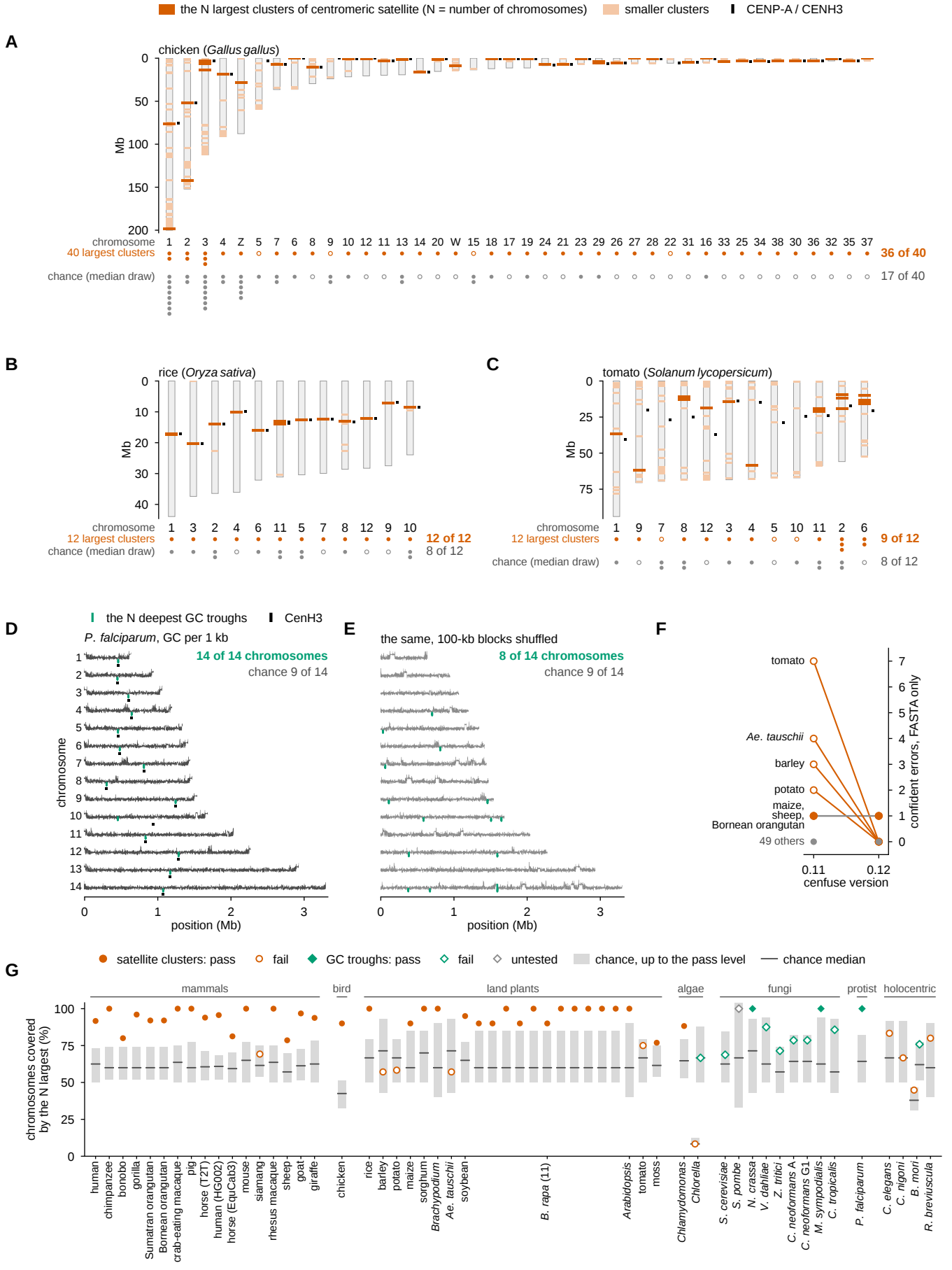

the N largest clusters of centromeric satellite (N = number of chromosomes)
smaller clusters
CENP-A / CENH3
A
chicken (Gallus gallus)
Mb
chromosome
40 largest clusters
chance (median draw)
36 of 40
17 of 40
B
rice (Oryza sativa)
12 largest clusters
12 of 12
8 of 12
C
tomato (Solanum lycopersicum)
9 of 12
8 of 12
D
the N deepest GC troughs
CenH3
P. falciparum, GC per 1 kb
14 of 14 chromosomes
chance 9 of 14
position (Mb)
E
the same, 100-kb blocks shuffled
8 of 14 chromosomes
chance 9 of 14
F
tomato
Ae. tauschii
barley
potato
maize, sheep, Bornean orangutan
49 others
confident errors, FASTA only
cenfuse version
G
satellite clusters: pass
fail
GC troughs: pass
fail
untested
chance, up to the pass level
chance median
chromosomes covered by the N largest (%)
mammals
bird
land plants
algae
fungi
protist
holocentric
human
chimpanzee
bonobo
gorilla
Sumatran orangutan
Bornean orangutan
crab-eating macaque
pig
horse (T2T)
human (HG002)
horse (EquCab3)
mouse
siamang
rhesus macaque
sheep
goat
giraffe
chicken
rice
barley
potato
maize
sorghum
Brachypodium
Ae. tauschii
soybean
B. rapa (11)
Arabidopsis
tomato
moss
Chlamydomonas
Chlorella
S. cerevisiae
S. pombe
N. crassa
V. dahliae
Z. tritici
C. neoformans A
C. neoformans G1
M. sympodialis
C. tropicalis
P. falciparum
C. elegans
C. nigoni
B. mori
R. breviuscula

A centromere occurs once per chromosome. So a genome's N largest centromeric candidates (N, its number of chromosomes) cover nearly every chromosome, whereas features that fall in proportion to length pile onto long chromosomes and leave others empty. The count recognizes centromeric features without any known centromere.

A–C, The largest centromeric satellite clusters of three genomes. In chicken they reach almost every chromosome, microchromosomes included, although the satellites differ from chromosome to chromosome; in rice one sits on every chromosome, on the centromere. In tomato they crowd onto some chromosomes and miss others, no better than chance: its centromeres lie elsewhere.

D, E, In *Plasmodium falciparum* the deepest AT-rich troughs fall one per chromosome, each on the centromere (**D**). Shuffling the genome in 100-kb blocks among chromosomes keeps every trough but breaks the rule: the deepest troughs crowd onto some chromosomes (**E**).

F, Where a genome's satellites fail the count, cenfuse 0.12 no longer calls their centromeres with confidence from the sequence alone, and the confident errors of tomato, potato, barley and *Aegilops tauschii* disappear.

G, Across every genome, the centromeric satellites of monocentric genomes pass, from primates to green algae; tomato, potato, barley, *Ae*. *tauschii*, siamang, *Chlorella* and the four holocentric genomes fail. AT troughs pass in *P*. *falciparum*, *Malassezia sympodialis* and *Neurospora crassa*, whose centromeres are AT-rich.

Terms. N: a genome's number of chromosomes. Clusters: arrays of the satellite families that cenfuse calls centromeric from the assembly alone (FASTA only), merged on each chromosome when less than 500 kb apart; size, their satellite bp. Covered: chromosomes holding at least one of the N largest clusters. Chance: the same count for N features placed on chromosomes in proportion to length, in 10,000 draws; P, the share of draws covering as many or more. Pass: $P \leq 0.05$; with 4 or fewer chromosomes even full coverage cannot pass (untested). Confident: high or medium confidence. CENP-A or CENH3 ChIP (truth) is used only to draw and score.

A–C, Chromosomes to scale, longest first, each genome to its own scale (Mb at left). Every cluster of the centromeric families is drawn (light), the N largest solid and widened; black bars: CENP-A (chicken; 33 of 40 chromosomes have data) or CENH3. Beneath, orange dots count the N largest clusters on each chromosome (open circle: none), gray dots those of one length-proportional draw, the first of the test's 10,000 draws to cover the median number of chromosomes. Chicken (13 centromeric families): 36 of 40 covered, chance 17 ($P \leq 0.0001$); 28 of the 40 clusters overlap CENP-A; only chromosomes 5, 9, 15 and 22 hold none, while chance would crowd 8 onto chromosome 1. Rice: 12 of 12, chance 8 ($P \leq 0.0001$); all 12 within 1 Mb of CENH3. Tomato: 9 of 12, chance 8 ($P = 0.24$); up to 3 clusters share a chromosome, 3 chromosomes hold none, and 1 of 12 lies within 1 Mb of CENH3: tomato centromeres are satellite-free, beside a dominant satellite.

D, E, GC of every 1-kb window along each *P. falciparum* chromosome (gray line; the test chooses the window, 1 to 500 kb, takes troughs at least 5 windows apart and corrects P for the 7 sizes tried). Green: the dips of the genome's 14 deepest troughs, pooled over chromosomes (GC 0.9–5.3%; genome median 18.9%); black bars: CenH3. **D,** 14 of 14 chromosomes covered, chance 9 ($P = 0.0007$); 13 of 14 troughs lie within 250 kb of CenH3 (chromosome 10: 482 kb away), and they become the FASTA-only calls (13 of 14 right, all low; 0 in 0.11). **E,** The control: 100-kb blocks of the GC profile reassigned at random among chromosomes, lengths kept (the first of the 20 stored shuffles of this genome): 8 of 14 covered at 1 kb, chance 9; it fails at every window ($P = 0.82$). Over 18 genomes, 16 of 360 shuffles pass (4.4%), the 5% the test allows.

F, Confident errors among FASTA-only calls, per genome, cenfuse 0.11 joined to 0.12 (open: satellite test fails; filled: passes). In the four genomes whose satellites fail, they fall from 16 to 0 (tomato 7, *Ae. tauschii* 4, barley 3, potato 2), while right calls stay (tomato 1 of 12, *Ae. tauschii* 1 of 7, barley 0 of 7, potato 2 of 11): the calls remain, at low confidence. The three remaining errors lie in genomes that pass. All 56 genomes: 19 to 3.

G, Every genome of the benchmark by lineage: chromosomes covered by the N largest clusters (circles) or troughs (diamonds), as a share of the chromosomes tested (*B. rapa*: 11 accessions). Gray: chance, from its 5th percentile up to the pass level, so filled marks lie above it; dash: chance median. Satellites pass in 36 of 46 genomes with a centromeric family and fail in 10; 10 have none. The GC test runs where most chromosomes lack a sequence call: it passes in *P. falciparum*, *M. sympodialis* (8 of 8, $P = 0.0054$) and *N. crassa* (7 of 7 at 200 kb, $P = 0.044$), fails in 8 and is untested in *Schizosaccharomyces pombe* (3 chromosomes).

Exceptions. Siamang fails ($P = 0.2$) although its centromeres are alpha-satellite arrays: on 8 chromosomes its arrays form two clusters more than 500 kb apart. *Chlamydomonas reinhardtii* passes (15 of 17, $P = 0.0006$) although its centromeres are LINE clusters, and so does moss (20 of 26, $P = 0.025$), whose centromeres are retrotransposon clusters; in both every call stays low, as no satellite family dominates (6 of 17 and 16 of 26 right). A family present on only some chromosomes caps the count. The test speaks about a genome, never about one locus, and at the 5% level one genome in 20 passes by chance.

Examples. A–C span the outcomes: a pass with chromosome-specific satellites (chicken), a plain pass (rice) and a failure (tomato); in each, the N largest clusters reproduce the stored test. *P. falciparum* has the strongest GC test and a CenH3 map; its shuffle is the first stored one, not a chosen one.

Analysis: asc-oulab:/data2/chris/centromere_benchmark/v2/v3/v4/v5/v6/v7/v8/v9/v10/v10.1/v10.2/v10.3/v10.4/v11 · Script: /data2/chris/centromere_benchmark/v2/v3/v4/v5/v6/v7/v8/v9/v10/v10.1/v10.2/v10.3/v10.4/v11/figures/fig_one_per_chromosome/fig_one_per_chromosome.py · 2026-11-07